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lated by IC 7805 (IC1). Capacitor C2 bypasses ripples present in the regulated supply. LED1 acts as the power indicator and R1 limits the current through LED1.

The heart of the system is microcontroller P89V51RD2 (IC2). The P89V51RD2 is an 8-bit 80C51 microcontroller with 64kB Flash and 1024 bytes of data RAM. It features three 16-bit timers/counters, programmable watchdog timer, eight interrupt sources with four priority levels, serial peripheral interface (SPI) and enhanced UART, programmable counter array with pulse-width modulation (PWM) and capture/compare functions, four 8-bit input/output ports with three high-current Port 1 pins. It has on-chip crystal oscillator and clock circuitry, which operates at a frequency of up to 40 MHz. It supports 12-clock per machine cycle (default) and 6-clock per machine cycle mode selected via software. The Flash program memory supports both parallel programming and in-system serial programming (ISP).

Data pins D0 through D7 of the LCD are connected to port pins P2.0 through P2.7 of microcontroller IC2, respectively. Control pins register-select (RS), read/write (R/W) and enable (EN) are connected to port pins P3.7, P3.6 and P3.5, respectively. Preset VR1 is connected to pin 3 of the LCD for contrast control. Power-on reset is provided by the combination of resistor R2 and capacitor C3. Switch S1 is used for manual reset. An 11.0592MHz crystal along with two 33pF capacitors provides the basic clock frequency to microcontroller IC2.

Port pins P1.0 and P1.1 of the microcontroller are used to control appliances like fan and bulb, respectively. The GSM module is connected to CON1 as shown in the circuit.

The GSM module is a specialised type of modem that accepts a SIM card and operates on the subscriber's mobile number over a network, just like a cellular phone. Basically, SIM300 is a triband GSM/GPRS engine that works on EGSM 900MHz, DCS 1800MHz and

PARTS LIST

Semiconductors:

IC1	- 7805, 5V regulator
IC2	- P89V51RD2 microcontroller
IC3	- MAX232 RS232 driver
T1, T2	- BC548 npn transistor
D1-D6	- 1N4007 rectifier diode
LED1	- 5mm LED
LCD	- 16x2 line LCD
GSM modem	- SIM300 V7.0 modem

Resistors (all 1/4-watt, ± 5 per cent carbon, otherwise stated):

R1	- 330-ohm
R2	- 10-kilo-ohm
R3, R4	- 2.2-kilo-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 0.1 μ F ceramic disk
C3, C6-C10	- 1 μ F, 16V electrolytic
C4, C5	- 33pF ceramic disk

Miscellaneous:

X1	- 230V AC primary to 12V, 500mA secondary transformer
S1	- Push-to-on tactile switch
RL1, RL2	- 12V, 1C/O relay
CON1	- DB9 COM port connector
X _{TAL}	- 11.0592 MHz

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

PCS 1900MHz frequencies. GSM modem is RS232-logic-level compatible. Here MAX232 is used to convert TTL into RS232 logic level and vice versa. Hence MAX232 is a voltage-level converter used between the microcontroller and the GSM module.

Working of the circuit is simple. To switch on the fan, send SMS 'fanon' from your mobile to the SIM number in the GSM modem. The GSM module receives your message and the microcontroller's port pin P1.0 goes high. This drives transistor T1 into saturation; relay RL1 energises and switches on the fan connected to it.

The fan can be turned off by sending SMS 'fanoff' from your mobile to the SIM number in the GSM modem. The GSM module receives your message and the microcontroller's port pin P1.0 goes low. Transistor T1 cuts off and relay RL1 de-energises to turn the fan 'off.'